

Cover Crops in Maize Rotations: Results of a Three-Season Field Study

Summary. We studied whether sowing a legume cover crop between maize seasons improves yields and soil quality on smallholder farms. Over three seasons, maize grown after a cover crop yielded more, needed less mineral fertiliser and had fewer weeds than maize grown after bare fallow. The extra labour for sowing and incorporating the cover crop was the main drawback reported by farmers.

Design. The study ran on twenty-four plots of 0.25 hectares on six farms in the same district, with four plots per farm. On each farm, two plots received a cover crop of velvet bean after the maize harvest and two plots were left as bare fallow, the usual practice in the area. Plots were assigned at random. Maize variety, planting dates and pest control followed the farmers' usual practice and were the same for all plots on a farm. Soil samples were taken before the first season and after the third.

Sowing and management. The cover crop was sown by hand within ten days after the harvest, while the soil still held enough moisture for the seeds to germinate. Sowing later than that led to poor establishment in two farms in the first season, which is why the window was kept strict afterwards. The cover crop was cut and left on the surface as mulch three weeks before maize planting, then worked lightly into the topsoil.

Yields. Averaged over the three seasons, maize yields rose by nine percent on plots with a cover crop compared with bare fallow plots. The difference was small in the first season and largest in the third, which suggests that the benefit builds up as soil organic matter increases. In the dry second season, cover crop plots also lost fewer plants to drought stress, probably because the mulch kept the soil moist for longer.

Fertiliser. Because velvet bean fixes nitrogen from the air, farmers on cover crop plots reduced nitrogen fertiliser by a quarter from the second season onwards without losing yield. This saving covered most of the cost of the cover crop seed. Phosphorus and potassium applications were not changed.

Soil. Soil organic carbon on cover crop plots increased by 0.3 percentage points over the three seasons, while it stayed level on bare fallow plots. Infiltration measured with a simple ring test was faster on cover crop plots, and farmers reported that the soil was easier to work.

Weeds. Weed biomass at the start of the maize season was forty percent lower on cover crop plots, because the cover crop and its mulch shaded out weeds during the dry period. Farmers needed one weeding round less per season on these plots.

Costs and labour. Sowing, cutting and incorporating the cover crop took about six extra hours of work per hectare each season. Several farmers said that this work falls at a time of year when labour is scarce, and three of the six farms said they would only continue if the seed were available locally.

Conclusions. A legume cover crop between maize seasons is worth recommending where seed is available and labour can be found shortly after the harvest. Extension services should

stress the short sowing window and the reduction in nitrogen fertiliser, which makes the practice pay for itself from the second season.